

Basic Profiling

Parallel and Heterogeneous Computing

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C Time Function

```
#include <time.h>
// ...
time_t t0 = time(NULL);
// ...
// do something
// ...
time_t t1 = time(NULL);
// duration (at a resolution of 1s)
time_t duration = t1 - t0;
```

- The time() function returns the number of seconds since 1 Jan 1970
- Only 1s resolution
- Don't
- Just don't

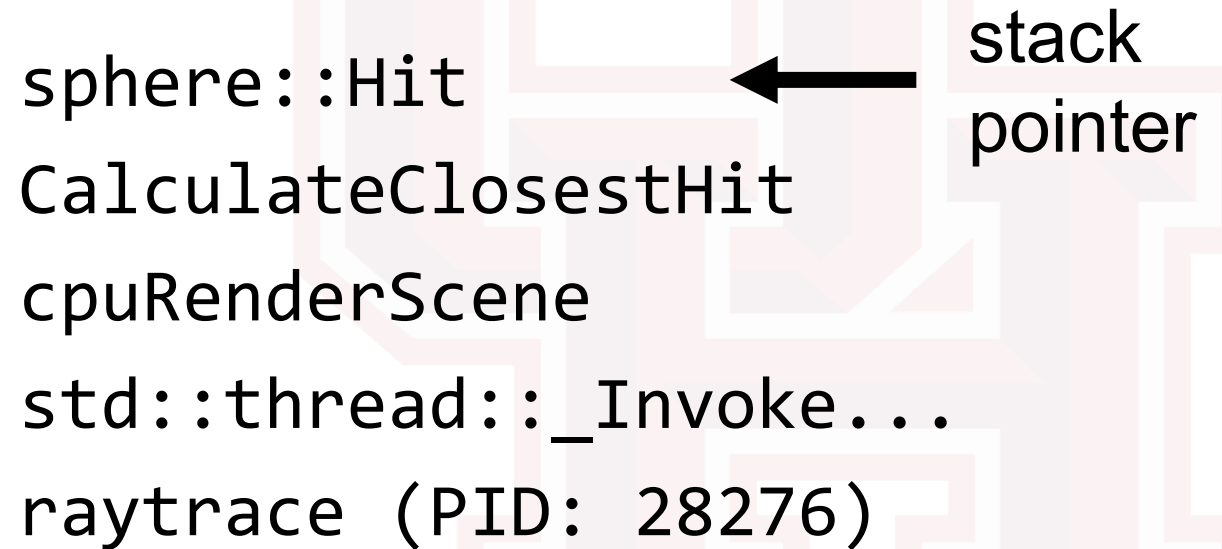
High Resolution Timers

```
#include <chrono>
// ...
auto t0 = std::chrono::high_resolution_clock::now();
// do something
auto t1 = std::chrono::high_resolution_clock::now();
std::chrono::duration<double> d = t1 - t0;
// get the time in seconds
// resolution is system dependent, but around 20 - 40 ns
double time_s = d.count();
```

Call Stack

- Keeps track of active subroutines in an executing program
- stack frame / activation record:
 - function parameters – values passed to the function
 - local variables – variables declared within the function
 - return address – instruction to return to
- stack pointer – points to the function location

- The stack pointer is “popped” when the function returns:

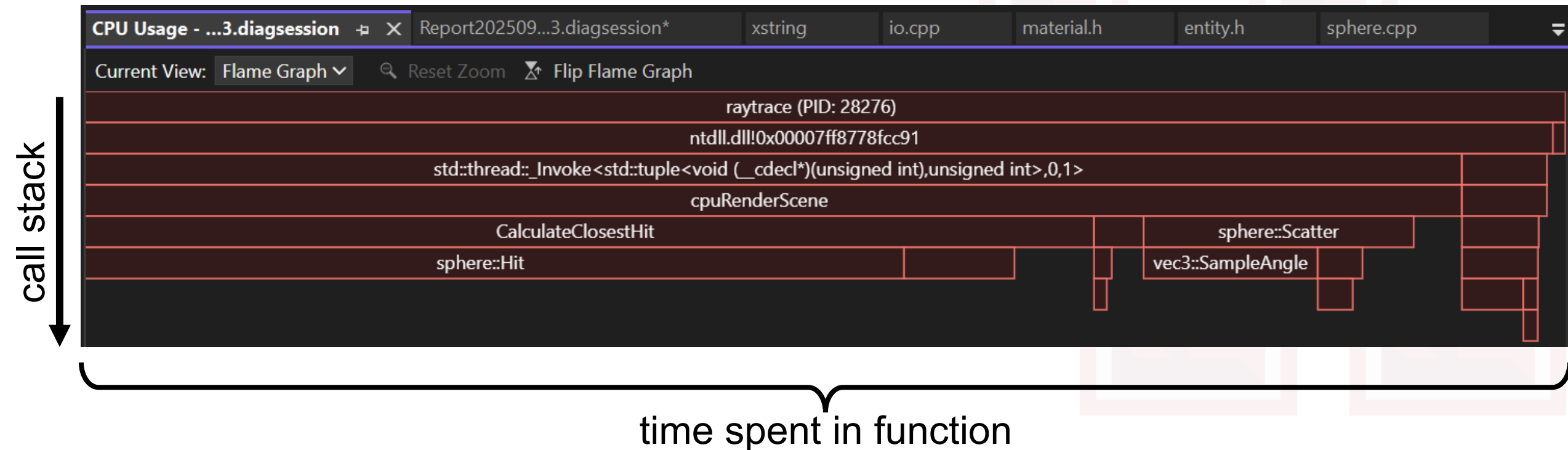


The diagram illustrates a call stack. It consists of a vertical list of function names: `sphere::Hit`, `CalculateClosestHit`, `cpuRenderScene`, `std::thread::_Invoke...`, and `raytrace (PID: 28276)`. An arrow labeled "stack pointer" points to the `sphere::Hit` entry, indicating the current function being executed. The stack grows downwards, with the most recent function at the top.

```
sphere::Hit
CalculateClosestHit
cpuRenderScene
std::thread::_Invoke...
raytrace (PID: 28276)
```

Flame Graph

- Profiling tools require compiling with debug symbols (but accurate results require optimization)
- Compilers usually provide an option (ex. RelWithDebInfo)



Profiling Demo

